

Development of a Mathematical Model for the Simulation of Ex Vivo Ocular Tissue Clearing

Proof of “Lipid degeneration rate equation” Equation

Proof. By assuming delipidation proceeds according to the second order rate law:

$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$

and by normalizing the final and initial lipid concentrations as ratios of the lipid concentration at that moment to the initial lipid concentration. For example, $[A]_t$ and $[A]_0$ would be represented as:

$$\begin{aligned} [A] &= \frac{C_{\text{lipid}}}{C_{\text{lipid},0}} \\ [A_0] &= \frac{C_{\text{lipid},0}}{C_{\text{lipid},0}} = 1 \end{aligned}$$

Substituting into the integrated rate law:

$$\begin{aligned} \frac{1}{[A]_t} &= kt + 1 \\ \Rightarrow [A]_t &= \frac{1}{1 + kt} \\ \Rightarrow \frac{C_{\text{lipid}}}{C_{\text{lipid},0}} &= \frac{1}{1 + kt} \end{aligned}$$

However, there exists a limitation to this equation. In order for the reaction to proceed, the amount of lipid reacted must be less than or equal to the concentration of CUBIC. To account for this boundary condition, the following equation was developed:

$$\boxed{\frac{C_{\text{lipid}}}{C_{\text{lipid},0}} = \frac{1}{kt + 1}, \quad 1 - \frac{C_{\text{lipid}}}{C_{\text{lipid},0}} \leq nC.}$$

Where n represents the reaction ratio of lipid concentration and CUBIC volume, and C represents the concentration of CUBIC. However, n can be negated if C is considered a ratio of filled pore space (i.e. $C = \frac{C_i}{\epsilon_0 V}$) instead of $C = C_i$.